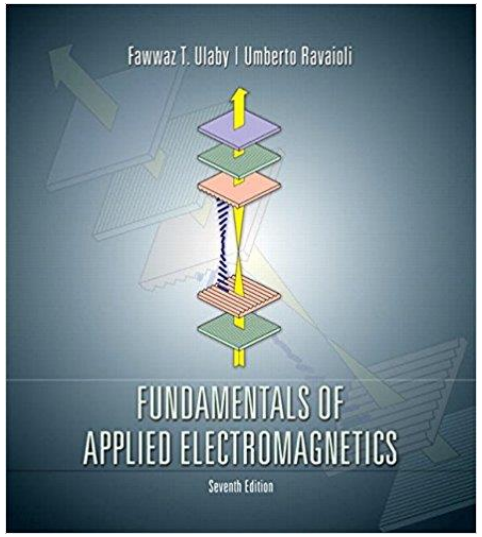
Reflection Coefficient & Standing Waves

ECE 370 – Electromagnetic Fields & Waves I

CSUN.

Course Announcements and Reading Material

- Assigned reading for today's lecture:
  - Sections 2.6 (~9 pages)
- Assigned reading for next lecture:
  - Sections 2.7-2.8 (~10 pages)



Lossless Transmission Lines

- Transmission lines can be designed to have low ohmic losses by selecting conductors with very high conductivities and dielectric materials with negligible conductivities
  - $R' \ll \omega L'$ and $G' \ll \omega C'$ which allow us to set $R' = G' \approx 0$

Lossless Transmission Lines

Transmission Line Parameters

	Propagation Constant $\gamma = \alpha + j\beta$	Phase Velocity u_p	Characteristic Impedance Z_0
General case	$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}$	$u_p = \omega/\beta$	$Z_0 = \sqrt{\dfrac{(R' + j\omega L')}{(G' + j\omega C')}}$
Lossless $(R' = G' = 0)$	$\alpha = 0,\ \beta = \omega\sqrt{\epsilon_r}/c$	$u_p = c/\sqrt{\epsilon_r}$	$Z_0 = \sqrt{L'/C'}$
Lossless coaxial	$\alpha = 0,\ \beta = \omega\sqrt{\epsilon_r}/c$	$u_p = c/\sqrt{\epsilon_r}$	$Z_0 = \left(60/\sqrt{\epsilon_r}\right)\ln(b/a)$
Lossless two-wire	$\alpha = 0,\ \beta = \omega\sqrt{\epsilon_r}/c$	$u_p = c/\sqrt{\epsilon_r}$	$Z_0 = \left(120/\sqrt{\epsilon_r}\right) \cdot \ln[(D/d) + \sqrt{(D/d)^2 - 1}]$ $Z_0 \approx \left(120/\sqrt{\epsilon_r}\right)\ln(2D/d)$, if $D \gg d$
Lossless parallel-plate	$\alpha = 0,\ \beta = \omega\sqrt{\epsilon_r}/c$	$u_p = c/\sqrt{\epsilon_r}$	$Z_0 = \left(120\pi/\sqrt{\epsilon_r}\right)(h/w)$

Notes: (1) $\mu = \mu_0$, $\epsilon = \epsilon_r\epsilon_0$, $c = 1/\sqrt{\mu_0\epsilon_0}$, and $\sqrt{\mu_0/\epsilon_0} \approx (120\pi)\ \Omega$, where ϵ_r is the relative permittivity of insulating material. (2) For coaxial line, a and b are radii of inner and outer conductors. (3) For two-wire line, d = wire diameter and D = separation between wire centers. (4) For parallel-plate line, w = width of plate and h = separation between the plates.

Lossless Transmission Lines

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z},$$

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}.$$

Waves travelling in the $+z$ direction

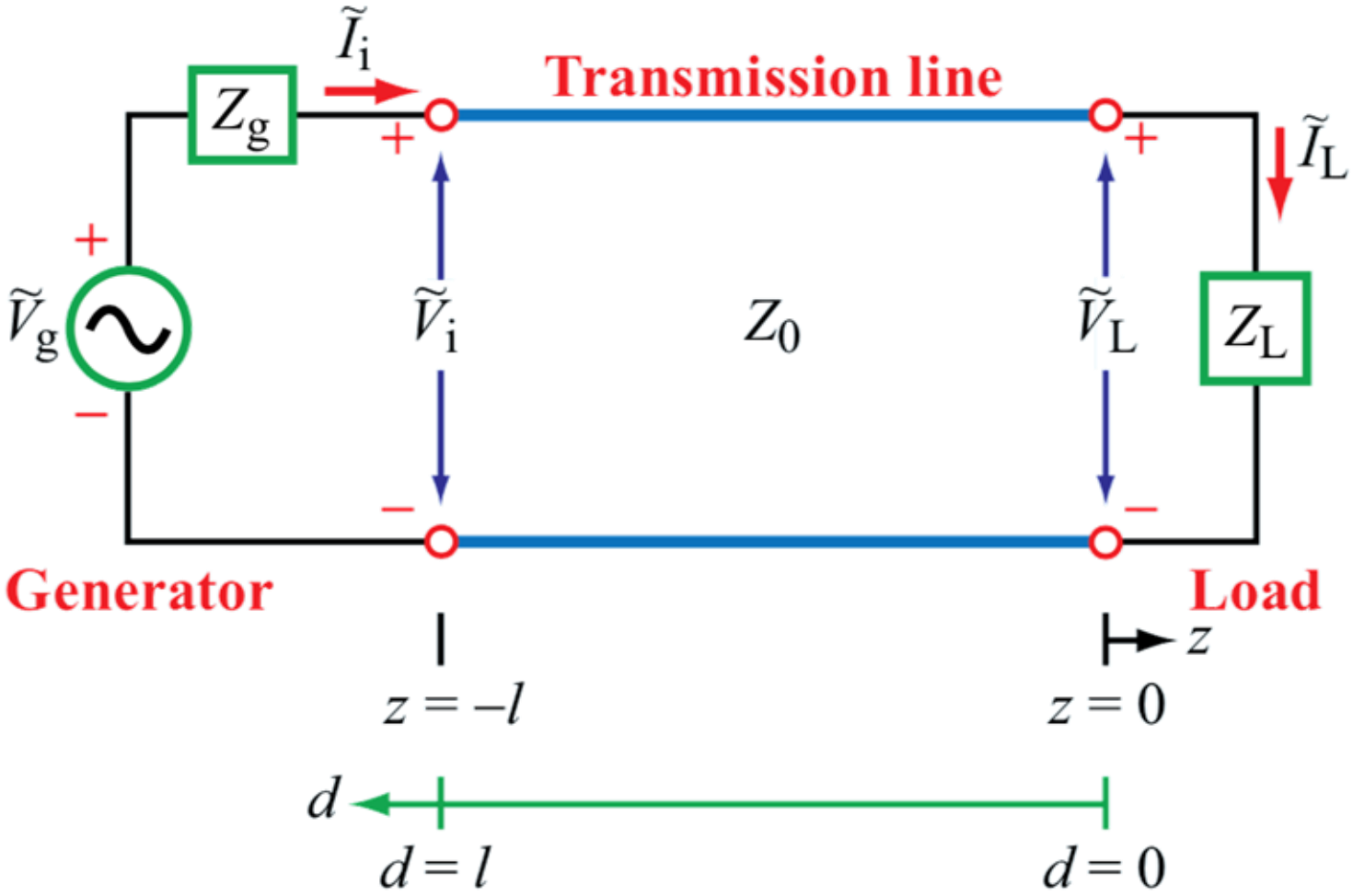
Waves travelling in the $-z$ direction

Incident wave

Reflected wave

- Note: the reference of the spatial coordinate z:
  - $z = 0$ corresponds to the location of the load
  - $z = -l$ corresponds to the location of the source
- For convenience (to avoid negatives) we define a distance from the load d where $d = -z$

Lossless Transmission Lines



Reflection Coefficient

- The ratio of the amplitudes of the reflected and incident voltage waves at the load is known as the voltage reflection coefficient Γ

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} = \frac{z_L - 1}{z_L + 1}$$

(dimensionless)

$$\frac{I_0^-}{I_0^+} = -\frac{V_0^-}{V_0^+} = -\Gamma$$

$$z_L = \frac{Z_L}{Z_0}$$

Normalized load impedance

Reflection Coefficient

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

- Z_L is in general a complex quantity, so Γ in general is also complex:

$$\Gamma = |\Gamma| e^{j\theta_r}$$

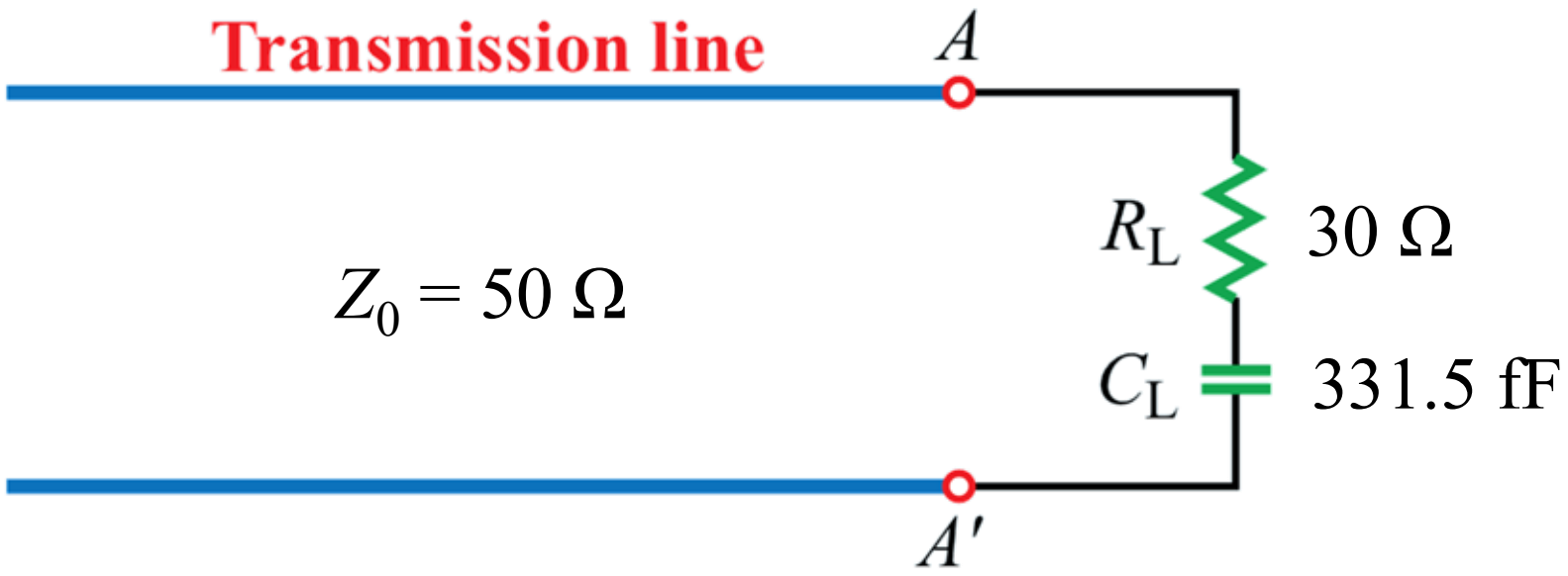
- A load is said to be matched to a transmission line if $Z_L = Z_0$ because then there will be no reflection by the load
  - $\Gamma = 0$ and $V_0^- = 0$
- If the load is an open circuit ($Z_L = \infty$): $\Gamma = 1$ and $V_0^- = V_0^+$
- If the load is a short circuit ($Z_L = 0$): $\Gamma = -1$ and $V_0^- = -V_0^+$

Reflection Coefficient $\Gamma = |\Gamma| e^{j\theta_r}$

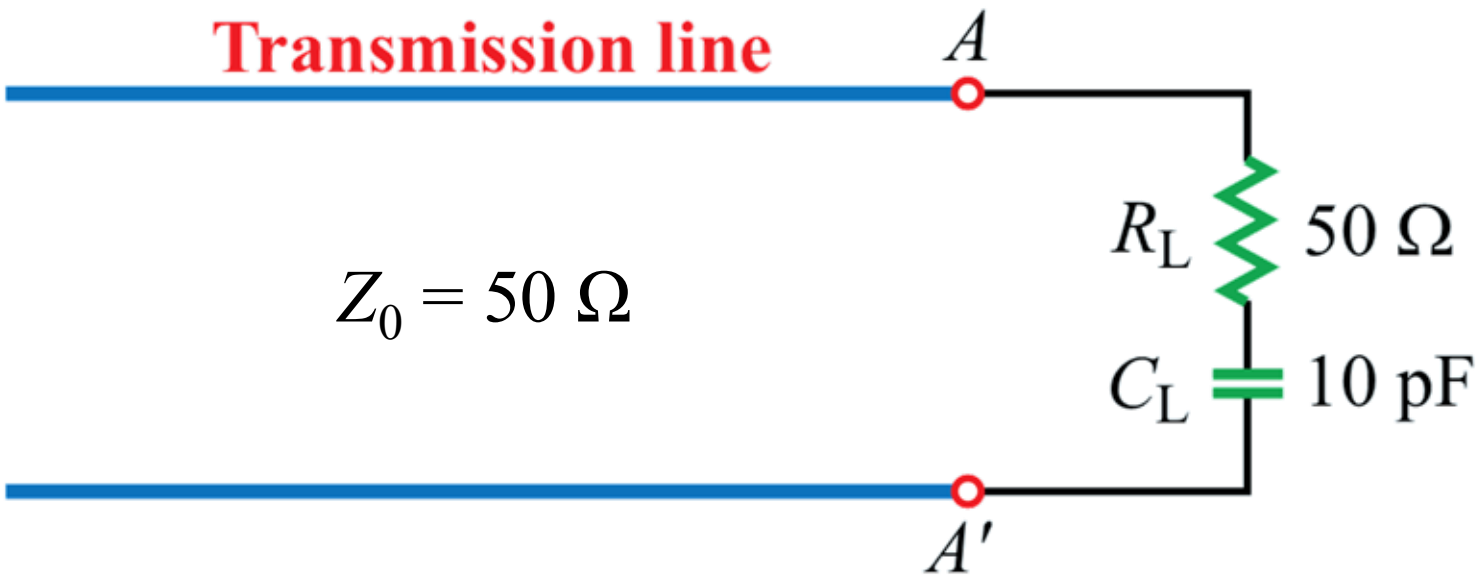
Load	$\lvert\Gamma\rvert$	θ_r
Z_0 $Z_L = (r + jx)Z_0$	$\left[\dfrac{(r-1)^2 + x^2}{(r+1)^2 + x^2}\right]^{1/2}$	$\tan^{-1}\left(\dfrac{x}{r-1}\right) - \tan^{-1}\left(\dfrac{x}{r+1}\right)$
Z_0 Z_0	0 (no reflection)	irrelevant
Z_0 (short)	1	$\pm 180°$ (phase opposition)
Z_0 (open)	1	0 (in-phase)
Z_0 $jX = j\omega L$	1	$\pm 180° - 2\tan^{-1} x$
Z_0 $jX = \dfrac{-j}{\omega C}$	1	$\pm 180° + 2\tan^{-1} x$

Reflection Coefficient Example

- Go through Examples 2-3 and 2-4 in text
- A 50 Ω transmission line is connected to a load consisting of a 30 Ω resistor in series with a 331.5 fF capacitor. Find the reflection coefficient at the load for a 2.4 GHz WiFi signal.



Reflection Coefficient Example



Standing Waves

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} \qquad V_0^- = \Gamma V_0^+$$

$$\widetilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \qquad \widetilde{V}(z) = V_0^+ (e^{-j\beta z} + \Gamma e^{j\beta z}),$$

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}. \qquad \tilde{I}(z) = \frac{V_0^+}{Z_0} (e^{-j\beta z} - \Gamma e^{j\beta z}).$$

$$|\widetilde{V}(z)| = \left\{ \left[V_0^+ (e^{-j\beta z} + |\Gamma| e^{j\theta_r} e^{j\beta z}) \right] \cdot \left[(V_0^+)^* (e^{j\beta z} + |\Gamma| e^{-j\theta_r} e^{-j\beta z}) \right] \right\}^{1/2}$$

$$= |V_0^+| \left[1 + |\Gamma|^2 + |\Gamma| (e^{j(2\beta z + \theta_r)} + e^{-j(2\beta z + \theta_r)}) \right]^{1/2}$$

$$= |V_0^+| \left[1 + |\Gamma|^2 + 2|\Gamma| \cos(2\beta z + \theta_r) \right]^{1/2}, \qquad e^{jx} + e^{-jx} = 2\cos x$$

Standing Waves

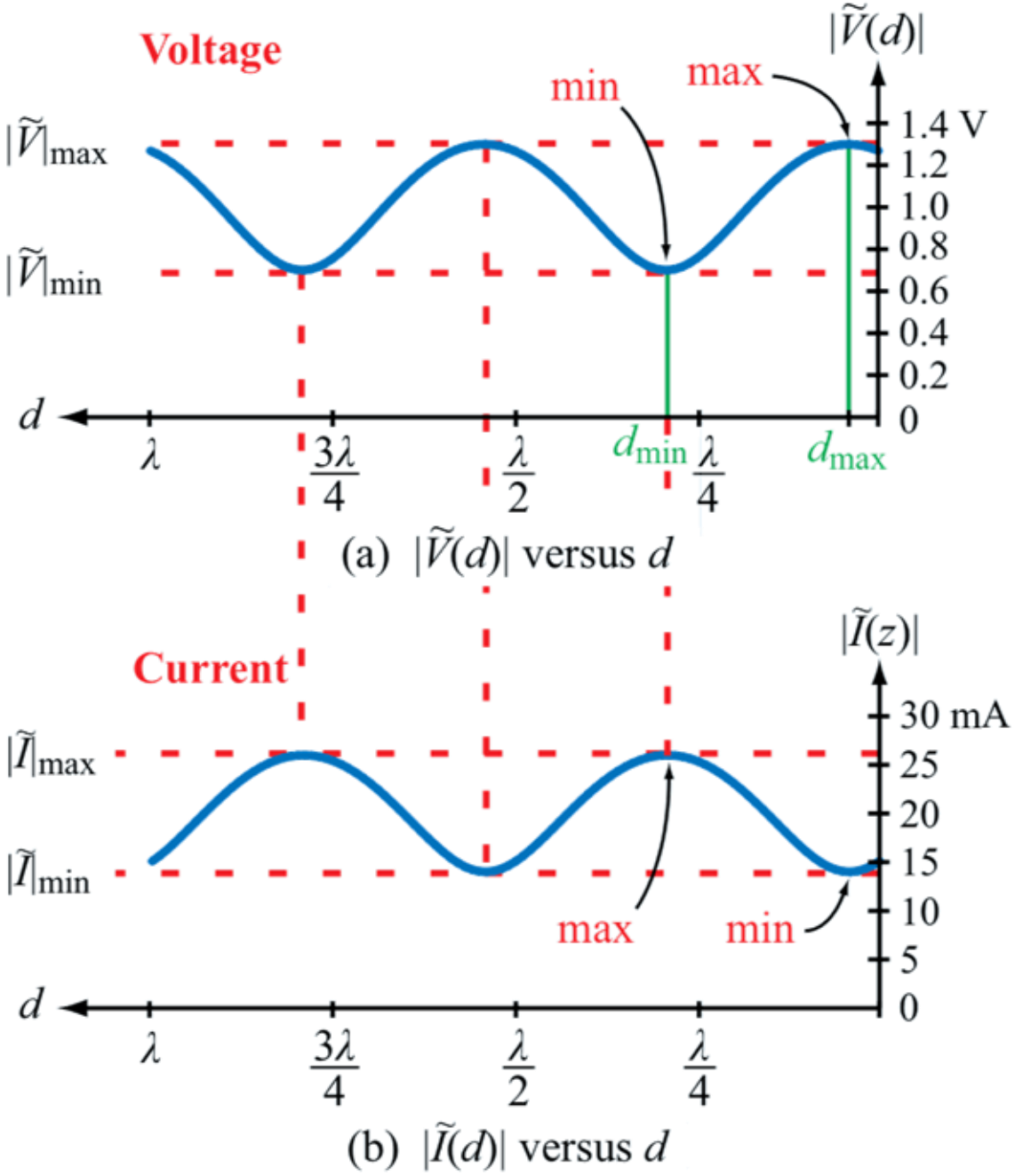
- Using d instead of z ($d = -z$),

$$|\tilde{V}(d)| = |V_0^+|\left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta d - \theta_r)\right]^{1/2}$$

$$|\tilde{I}(d)| = \frac{|V_0^+|}{Z_0}\left[1 + |\Gamma|^2 - 2|\Gamma|\cos(2\beta d - \theta_r)\right]^{1/2}$$

- Standing-wave pattern for a lossless transmission line with $Z_0 = 50\ \Omega$, terminated in a load with a reflection coefficient $\Gamma = 0.3e^{j\,30°}$ and $|V_0^+| = 1$ V
  - The standing wave ratio is:

$$S = |\tilde{V}|_{\max}/|\tilde{V}|_{\min} = 1.3/0.7 = 1.86.$$



Standing Waves

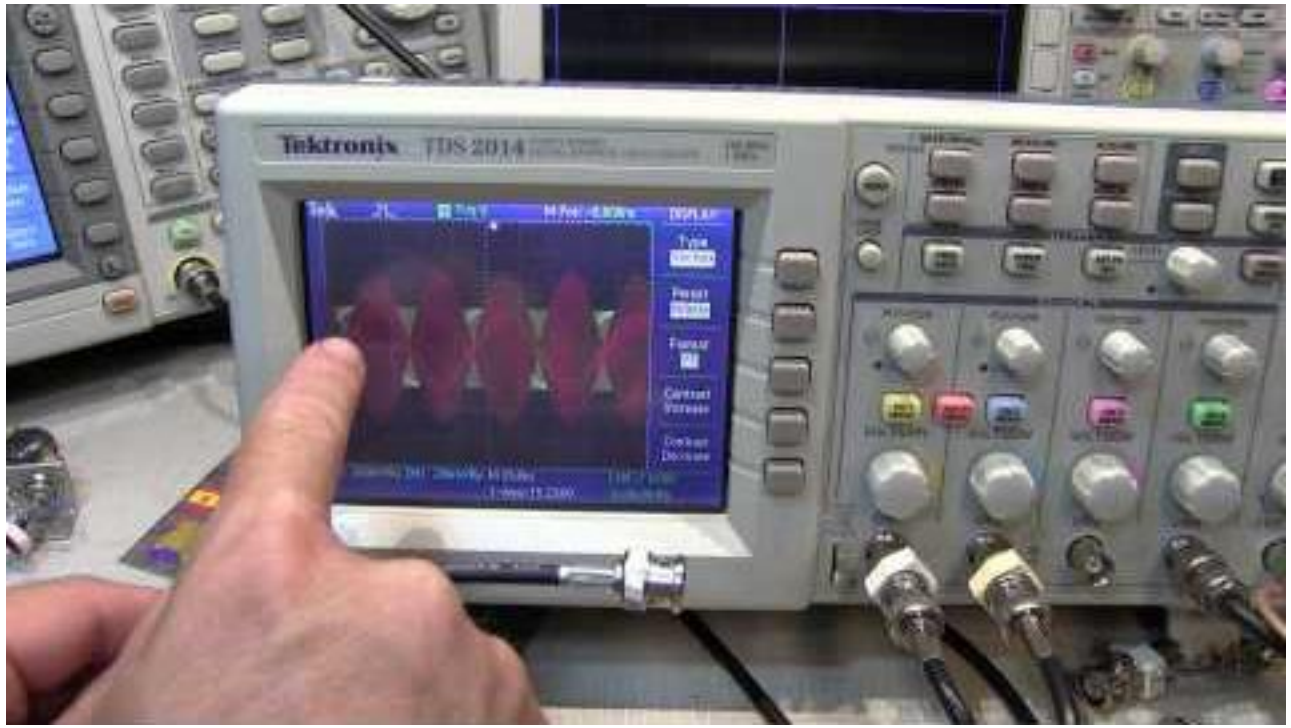
- The sinusoidal patterns are called standing waves and are caused by the interference of the two traveling waves
  - The maximum value of $|\tilde{V}(d)|$ occurs where the incident and reflected waves are in-phase $(2\beta d - \theta_r = 2n\pi)$
  - The minimum value of $|\tilde{V}(d)|$ occurs when the two waves are out-of-phase $(2\beta d - \theta_r = (2n+1)\pi)$
- With no reflected wave (i.e., the matched case) there is no interference and there are no standing waves
- If we observe the variation of the instantaneous voltage as a function of time at location $d = d_{\max}$, the variation of the $|\tilde{V}(d)|$ would be $\cos \omega t$
  - The voltage oscillates from 1.3 V and -1.3 V on the previous plot at $d_{\max}$

Visualizing Standing Waves

https://www.youtube.com/watch?feature=oembed&v=-gr7KmTOrx0

https://www.youtube.com/watch?v=NpEevfOU4Z8

Visualizing Standing Waves



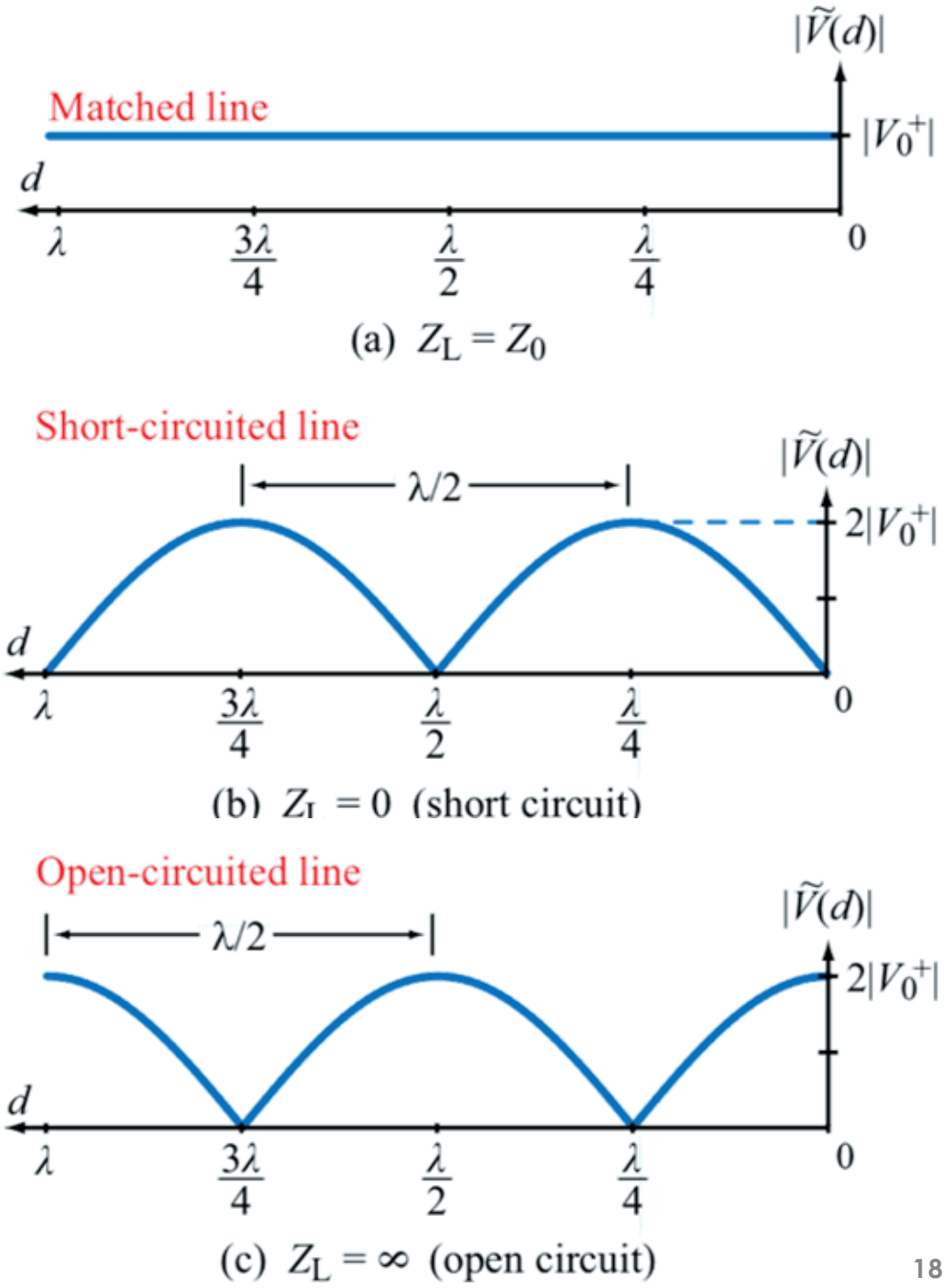
https://www.youtube.com/watch?v=M1PgCOTDjvI

Standing Waves

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$|\tilde{V}(d)| = |V_0^+| \left[1 + |\Gamma|^2 + 2|\Gamma| \cos(2\beta d - \theta_r)\right]^{1/2}$$

- Matched line: $\Gamma = 0$
- Short-circuited line: $\Gamma = -1$
- Open-circuited line: $\Gamma = 1$



Standing Waves

$$|\tilde{V}(d)| = |V_0^+|\left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta d - \theta_r)\right]^{1/2}$$

- At the distance from the load at which $|\tilde{V}(d)|$ is a maximum, d_{max}:

$$|\tilde{V}(d)| = |\tilde{V}|_{max} = |V_0^+|[1 + |\Gamma|]$$

- This occurs when $2\beta d_{max} - \theta_r = 2n\pi$
  - where $n = 0$ or a positive integer
- Solving for d_{max} gives

$$d_{max} = \frac{\theta_r + 2n\pi}{2\beta} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2},$$

$$\begin{cases} n = 1, 2, \ldots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \ldots & \text{if } \theta_r \geq 0, \end{cases}$$

$$\beta = 2\pi/\lambda$$

Standing Waves

$$|\tilde{V}(d)| = |V_0^+|\left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta d - \theta_r)\right]^{1/2}$$

- At the distance from the load at which $|\tilde{V}(d)|$ is a minimum, $d_{\min}$:

$$|\tilde{V}|_{\min} = |V_0^+|[1 - |\Gamma|]$$

- This occurs when $(2\beta d_{\min} - \theta_r) = (2n + 1)\pi$
  - The first minimum corresponds to $n = 0$
- The spacing between a maximum $d_{\max}$ and the adjacent minimum $d_{\min}$ is $\lambda/4$. Hence, the first minimum occurs at

$$d_{\min} = \begin{cases} d_{\max} + \lambda/4, & \text{if } d_{\max} < \lambda/4, \\ d_{\max} - \lambda/4, & \text{if } d_{\max} \geq \lambda/4. \end{cases}$$

  - The locations on the line corresponding to voltage maxima correspond to current minima, and vice versa

Standing Waves

- The ratio of $|\tilde{V}|_{\max}$ to $|\tilde{V}|_{\min}$ is called the voltage standing-wave ratio, S

$$S = \frac{|\tilde{V}|_{\max}}{|\tilde{V}|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad \textbf{(dimensionless)}$$

- Also commonly referred to as VSWR
- It provides a measure of the mismatch between the load and the transmission line
  - For a matched load with $\Gamma = 0$, $S = 1$
  - For a line with $|\Gamma| = 1$, $S = \infty$

Standing Wave Ratio Examples

- Go through Example 2-5 and 2-6
- A $140\ \Omega$ lossless line is terminated in a load impedance $Z_L = (280 + j182)\ \Omega$. If $\lambda = 72$ cm, find
  a) The reflection coefficient Γ
  b) The voltage standing-wave ratio S
  c) The locations of voltage maxima
  d) The locations of voltage minima

Standing Wave Ratio Examples